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1	1	1	1	1	1	1
2	2	2	2	2	2	2
3	3	3	3	3	3	3
4	4	4	4	4	4	4
5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

The OMR Sheet will be computer checked. Fill the circles completely and dark enough for proper detection. Use Dark HB pencil or ballpen (black or blue) for marking. While erasing, erase properly.

Avoid Improper Marking



Partially Filled



Lightly filled



Analysis & Design of Algorithms 2026 — Quiz II

Duration: 1 hour

Total Marks: 20

Name: _____

Roll No.: _____

Problem 1. You are given n points labeled $1, 2, \dots, n$ on a line and a collection of k intervals

$$I_1, I_2, \dots, I_k,$$

where each interval $I_i = [s_i, e_i]$ satisfies $1 \leq s_i \leq e_i \leq n$ and s_i, e_i are *integers*. A subset of points $X \subseteq \{1, 2, \dots, n\}$ is said to *hit* an interval I_i if

$$X \cap I_i \neq \emptyset.$$

The goal is to find a hitting set of minimum cardinality. Consider the following greedy algorithm.

1. Sort the intervals in non-decreasing order of their right endpoints.
2. Initialize $X \leftarrow \emptyset$.
3. Process the intervals in this order. For an interval $[s, e]$, if it is not already hit by a point in X , add the point e to X .

The algorithm outputs the set X . Prove that the algorithm produces a hitting set of minimum possible cardinality. Your proof should use one of the rigorous methods like exchange arguments, inductions, lower bounds on opt etc. Any verbose, hand-wavy arguments will fetch zero credit.

Solution. Let:

- $G = \{g_1, g_2, \dots, g_m\}$ be the points selected by the greedy algorithm, in the order they are added.
- $O = \{o_1, o_2, \dots, o_\ell\}$ be the points selected by an optimal solution, sorted from left to right.

Rubric: +2 for starting the proof setup correctly as above with all details like ‘sorted from left to right’. We prove that $m = \ell$, i.e., the greedy algorithm is optimal.

Step 1: Greedy Stays Ahead Lemma **Rubric:** +4 in total for this lemma, detail breakup below. For every $j \geq 1$, if both greedy and optimal selects j points, then

$$g_j \geq o_j.$$

Rubric: +1 for stating the above induction hypothesis properly.

We proceed by induction on j .

Base case ($j = 1$).

Let $I = [s, e]$ be the first interval processed by the greedy algorithm. Greedy selects $g_1 = e$, the smallest right endpoint among all intervals.

Any feasible solution, in particular the optimal one, must place at least one point inside I . Hence,

$$o_1 \in [s, e] \Rightarrow o_1 \leq e = g_1.$$

Thus, $g_1 \geq o_1$. **Rubric:** +0.5 for stating the base case, +0.5 for proving the base case.

Inductive step. Assume that for some $j \geq 1$,

$$g_i \geq o_i \quad \text{for all } i \leq j.$$

Consider the moment when greedy selects g_{j+1} . Let $I = [s, e]$ be the interval that triggers this selection. By definition:

- I is not hit by $\{g_1, \dots, g_j\}$, and

- greedy sets $g_{j+1} = e$.

Since $g_i \geq o_i$ for all $i \leq j$, none of the points $o_1, \dots, o_j$ can hit I either. Therefore, the optimal solution must use some point o_{j+1} to hit I .

Thus,

$$o_{j+1} \in [s, e] \Rightarrow o_{j+1} \leq e = g_{j+1}.$$

Hence $g_{j+1} \geq o_{j+1}$, completing the induction.

Rubric: +2 for carrying out the induction step correctly. Evaluation is somewhat subjective on how correct and complete the proof is.

Step 2: Greedy Cannot Use More Points Than OPT. Assume for contradiction that the greedy algorithm selects more points than the optimal solution, i.e., $m > \ell$. **Rubric:** +1 for stating this correctly

Since greedy selects a point with index $\ell + 1$, it must have encountered an interval $I = [s, e]$ such that $s > g_\ell \geq o_\ell$, where the second inequality follows from the Greedy Stays Ahead Lemma. But then the optimal solution did not hit I which is a contradiction.

Rubric: +3 for correctly writing the proof above. Again, evaluation will be subjective.

Problem 2. Let $c > 1$ be a fixed integer. You are given an unlimited supply of coins with denominations

$$\{1, c, c^2, \dots, c^k\}.$$

Given a target integer amount L , the goal is to determine the *smallest* possible collection of coins that add up to L . Consider the following greedy algorithm:

1. While the remaining amount is positive, select a coin of the largest denomination that does not exceed the remaining amount.
2. Subtract the value of the selected coin from the remaining amount.

Prove that the greedy algorithm stated above produces the optimal solution to the coin change problem. Your solution must use an exchange argument.

Solution. We will prove this by an exchange argument. Suppose we order the optimal solution in non-decreasing order of their denominations. Greedy, by definition, picks the coins in this order. **Rubric:** +2 for saying this with all details correct.

Now assume that the greedy and optimal solution differs, for the first time in the above ordering at some position where greedy uses c^j but optimal uses c^ℓ for some $\ell < j$.

Rubric: +1 for stating this properly.

Now we claim that *there exists* a denomination $k \leq \ell$ such that optimal uses c coins of that denomination.

Rubric: +1 for stating this.

Suppose the optimal solution uses all denominations $k \leq \ell$ strictly less than c times. But then the total that the optimal can obtain from these coins is at most

$$(c-1)(c^\ell + c^{\ell-1} + \dots + 1) = (c-1) \frac{c^{\ell+1} - 1}{c-1} = c^{\ell+1} - 1 < c^j$$

This is a contradiction since greedy had used a coin of denomination c^j which optimal is not able to cover for.

Rubric: +3 for the above proof. If you proved this assuming $k = j - 1$, you get 0.5 point deducted.

Finally the exchange argument. From the above, we know that there is at least one denomination $k < j$ which has been used c times in optimal. But then we can replace all those coins (or exchange) with just one coin of denomination $c^k + 1$. But this means optimal is not using the minimal possible number of coins again leading to a contradiction.

Rubric: +3 for finishing off the proof with the above (or similar) exchange argument.

Remark; You cannot directly use facts like ‘greedy is exactly same as base- c representation of the number L etc.’ I mean, you can. But then you have to show that this is the optimal way of representing any number in base c etc. In fact, this is exactly what you have to prove as above.